SIMATS ENGINEERING
ASSESSMENT TOOL 3: Class Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: ***Apply image enhancement and segmentation techniques for extracting meaningful regions from images.*** **(BTL 3)**

Assessment Tool Weightage: 10%

Code of Conduct:

I _Bharath Sankar (192425046)_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

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Signature of the student:

Rubrics for Evaluation

Criteria	*Weightage*	*Level 4 – Exemplary*	*Level 3 – Proficient*	*Level 2 – Developing*	*Level 1 – Beginning*	*Marks(100)*
Understanding of Concepts	*25%*	*Demonstrates thorough, accurate understanding of concepts*	*Good understanding with minor gaps*	*Basic understanding with some misconceptions*	*Poor understanding; major misconceptions*	
Explanation	*25%*	*Detailed, well-explained answers with relevant examples*	*Clear explanation with adequate details*	*Limited explanation; lacks depth*	*Poor or unclear explanation*	
Application	*20%*	*Effectively applies concepts with strong analytical ability*	*Applies concepts with minor errors*	*Limited application with weak analysis*	*No application or incorrect analysis*	
Organization	*15%*	*Well-structured answers with logical flow and coherence*	*Organized with minor issues in flow*	*Basic structure, lacks clarity*	*Poor organization, incoherent answers*	

Accuracy	***10%***	***Completely accurate and covers all required points***	***Mostly accurate with minor omissions***	***Partially correct with missing points***	***Incorrect answers with major gaps***	
Clarity	***5%***	***Clear, neat, professional presentation***	***Understandable with minor issues***	***Limited clarity, readability issues***	***Poorly written, difficult to understand***	

1.An image appears too bright.
Explain and justify the most suitable grey level transformation to improve visibility, clearly describing the concept, application process, and expected outcome.

A bright image contains many high-intensity pixel values, making details difficult to see. The most suitable grey level transformation is **Negative Transformation** (or brightness reduction).

- **Concept:** Negative transformation inverts the intensity values.
- **Formula:** $s = 255 - r$, where *r* is the input pixel value.
- **Working:** Bright pixels become dark, and dark pixels become bright.
- **Application:** It is useful for improving visibility in overexposed images and highlighting hidden details.
- **Expected Outcome:** The image becomes balanced, and details in bright regions become clearly visible.

2.A dark image lacks detail.
Describe and justify an appropriate grey level transformation to enhance visibility, explaining how it improves low-intensity regions with proper reasoning.

A dark image has low-intensity pixels that hide important information. The suitable transformation is **Log Transformation**.

- **Concept:** Log transformation expands low-intensity values more than high-intensity values.
- **Formula:** $s = c\log(1 + r)$
- **Working:** Dark pixels become brighter while bright pixels change only slightly.
- **Application:** Used in medical and satellite images to reveal hidden details.

- **Expected Outcome:** The image becomes brighter, and details in dark regions are clearly visible.

3.An image has poor contrast.
Explain a suitable histogram-based technique for contrast enhancement and illustrate how it improves image quality with logical reasoning.

The best histogram-based technique is **Histogram Equalization**.

- **Concept:** It redistributes the intensity values across the entire range.
- **Working:** Pixel values are spread uniformly from dark to bright.
- **Application:** Used to improve low-contrast photographs and medical images.
- **Advantage:** Enhances visibility of objects and edges.
- **Expected Outcome:** The image has improved contrast and better visual quality.

4A low-contrast image requires uniform intensity distribution.
Describe and justify the method you would apply, explaining the concept, working process, and its effect on intensity levels.

The suitable method is **Histogram Equalization**.

- **Concept:** It converts the histogram into a more uniform distribution.
- **Working:** Frequently occurring intensity values are spread over the available range.
- **Application:** Improves global contrast in grayscale images.
- **Benefit:** Makes hidden features more visible.
- **Expected Outcome:** Uniform intensity distribution with enhanced image clarity.

5.An image shows uneven brightness across regions.
Explain and apply an appropriate enhancement technique to correct non-uniform illumination, providing a structured and accurate explanation.

The appropriate enhancement technique is **Adaptive Histogram Equalization (CLAHE).**

- **Concept:** The image is divided into small regions, and histogram equalization is applied separately.

- **Working:** Each region is enhanced independently while limiting excessive contrast.
- **Application:** Used for images with uneven lighting.
- **Advantage:** Corrects local brightness without amplifying noise.
- **Expected Outcome:** Uniform brightness and improved visibility throughout the image.

6.A noisy image contains random intensity variations.
Explain and justify the most suitable smoothing filter, including its working mechanism and impact on noise reduction.

The most suitable smoothing filter is the **Median Filter**.

- **Concept:** It replaces each pixel with the median value of neighboring pixels.
- **Working:** Removes random noise without averaging pixel values.
- **Application:** Highly effective for salt-and-pepper noise.
- **Advantage:** Preserves image edges better than mean filtering.
- **Expected Outcome:** Noise is reduced while edges remain sharp.

7.An image requires noise reduction without significant loss of details.
Describe and apply an appropriate filtering approach, explaining how it balances noise removal and detail preservation.

The suitable filtering approach is the **Bilateral Filter**.

- **Concept:** It smooths the image while preserving edges.
- **Working:** Uses both spatial distance and pixel intensity differences.
- **Application:** Used in photography and computer vision.
- **Advantage:** Reduces noise without blurring important edges.
- **Expected Outcome:** A smooth image with preserved fine details.

8.An image appears blurred after smoothing.
Explain and justify a suitable sharpening method to restore details, including its working principle and expected results.

The suitable sharpening method is **Laplacian Sharpening**.

- **Concept:** It uses the second derivative to detect edges.
- **Working:** High-frequency components are added back to the image.
- **Application:** Restores sharpness after smoothing.
- **Advantage:** Enhances edges and object boundaries.
- **Expected Outcome:** The image becomes sharper with improved detail.

9.Fine details in an image are not clearly visible.
Describe and justify the filtering technique used to enhance details, explaining its effect on high-frequency components.

The appropriate filtering technique is the **High-Pass Filter**.

- **Concept:** High-pass filtering enhances high-frequency components.
- **Working:** It suppresses low-frequency background information and emphasizes edges.
- **Application:** Used in image sharpening and edge enhancement.
- **Advantage:** Makes textures and fine details more visible.
- **Expected Outcome:** Clearer edges and enhanced image details.

10A noisy image must be preprocessed before further analysis.
Explain and apply a suitable spatial filtering method, ensuring a clear, structured explanation with justification.

The suitable spatial filtering method is the **Gaussian Filter**.

- **Concept:** It smooths the image using a Gaussian kernel.
- **Working:** Neighboring pixels are weighted according to the Gaussian function.
- **Application:** Used as a preprocessing step before segmentation and feature extraction.
- **Advantage:** Reduces Gaussian noise while maintaining overall image quality.
- **Expected Outcome:** A cleaner image that improves the accuracy of further image processing tasks.

These answers are based on the topics and questions in your uploaded assessment document.